Module-12c Linear filtering with DFT

During the course of our study we have found that the output of an LTI system with impulse response $h[n]$ is given by convolution of $h[n]$ with the input $x[n]$, i.e., $y[n] = x[n] * h[n]$. Recall that if we consider the DTFT of $h[n]$ and $x[n]$ to be $H(e^{j\omega})$ and $X(e^{j\omega})$, respectively, then the frequency response of the output is given by $Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$. Thus, an LTI system changes the frequency spectrum of the input. This process is also known in the literature as *linear filtering*. This is what we have been doing from almost the start of our classes. We are just giving it a name now – linear filtering. In short, Linear filtering is a process in which part of the signal frequency spectrum is modified by the transfer function of the filter (LTI system).

Note that computing $H(e^{j\omega})$ and $X(e^{j\omega})$ is not possible in a computer, because these are continuous functions of ω. One idea can be that we sample $X(e^{j\omega})$ and $H(e^{j\omega})$, i.e., find the DFT of $x[n]$ and $h[n]$, respectively at N-equally frequencies. Let the DFTs of $x[n]$ and $h[n]$ be $X[k]$ and $H[k]$. Note that IDFT of $X[k]H[k]$ will give us the circular convolution of $x[n]$ and $h[n]$, i.e., $x[n] \otimes h[n]$. However, the output $y[n]$ is the linear covolution of $x[n]$ and $y[n]$. So, IDFT of $X[k]H[k]$ would not help us in getting the output. So, the straight forward question is can we use DFT to compute linear convolution. Let us explore this.

1 Linear filtering

Suppose that we have a finite-duration sequence $x[n]$ of length L and the impulse response of the system is of length M. Without loss of generality, let

$$x[n] = 0, \qquad n < 0 \text{ and } n \geqslant L, \qquad h[n] = 0, \qquad n < 0 \text{ and } m \geqslant M.$$

The output sequence is then given by

$$y[n] = h[n] * x[n] = \sum_{k=0}^{M-1} h[k]x[n-k]. \tag{1}$$

Since $h[n]$ and $x[n]$ are finite-duration sequences, their convolution is also finite-duration. Further, the duration of $y[n]$ is $L + M - 1$ (Recall you established this relation in Problem 6a of Assignment-1).

In the frequency domain, we have

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega}). \tag{2}$$

If the $(L+M-1)$ duration sequence $y[n]$ has to be uniquely represented in the frequency domain by samples of $Y(e^{j\omega})$, then the number of samples must equal or exceed $L+M-1$. Therefore, a DFT of size $N \geqslant L+M-1$, is required to represent $y[n]$ in the frequency domain (see Frequeny Domain Sampling Module-12a for minimum number of samples required for perfect reconstruction). Now for $k = 0, 1, \cdots, N-1$, $Y[k] = Y(e^{j\omega})|_{\omega=\frac{2\pi}{N}k} = X(e^{j\omega})H(e^{j\omega})|_{\omega=\frac{2\pi}{N}k}$. Then

$$Y[k] = X[k]H[k] \qquad k = 0, 1, \cdots, N-1, \tag{3}$$

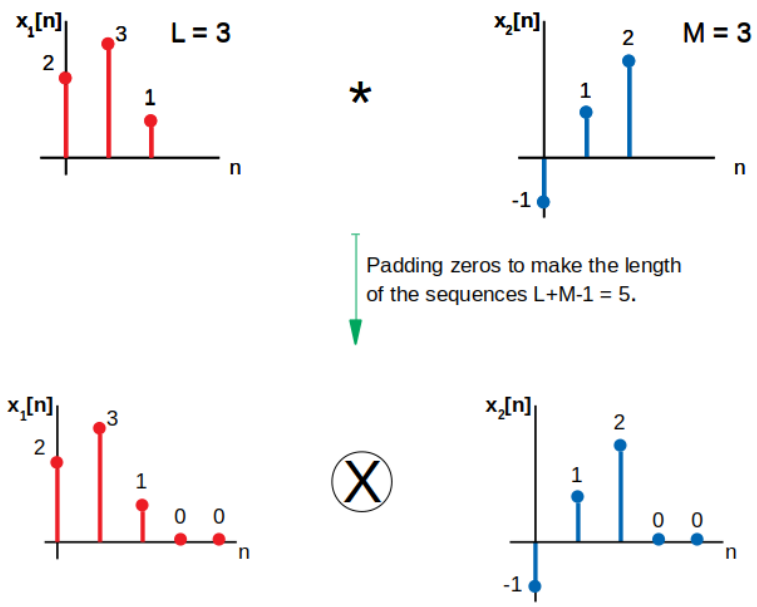
where $X[k]$ and $Y[k]$ are N-point DFTs of the corresponding sequences $x[n]$ and $h[n]$ with $N \geqslant L+M-1$. Since the sequences $x[n]$ and $h[n]$ have a duration less than N, we simply pad these sequences with zeros to increase their length to N. This increase in the size of the sequences $x[n]$ and $h[n]$ does not alter their spectra $X(e^{j\omega})$ and $Y(e^{j\omega})$.

Note that in order to uniquely reconstruct $x[n]$ from $X(e^{j\omega})$, we can sample $X(e^{j\omega})$ at the frequencies $\omega_k = \frac{2\pi}{L}$. However, now we are sampling $X(e^{j\omega})$ at $\Omega_k = \frac{2\pi}{N}$. Since $N \geq L$, we have more number of samples of $X(e^{j\omega})$ if we sample at Ω_k compared to sampling at ω_k. A similar argument holds for $H(e^{j\omega})$ as well, since $N \geq M$.

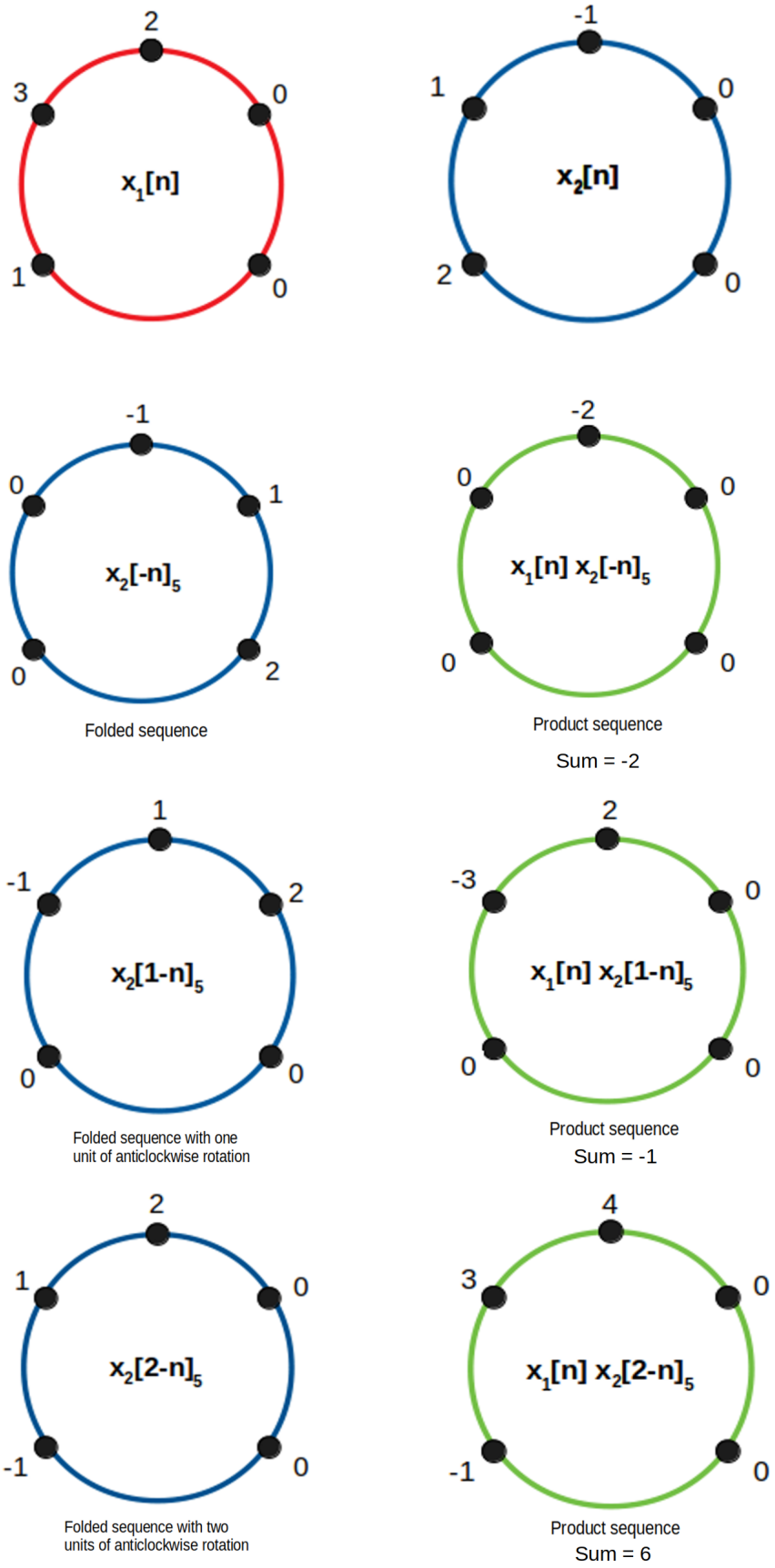
Since $N = (L+M-1)$-point DFT of $y[n]$ is sufficient to represent $y[n]$ in the frequency domain, it follows that the multiplication of the N-point DFTs $X[k]$ and $H[k]$, by equation (3), followed by computation of N-point IDFT must yield the sequence $y[n]$.

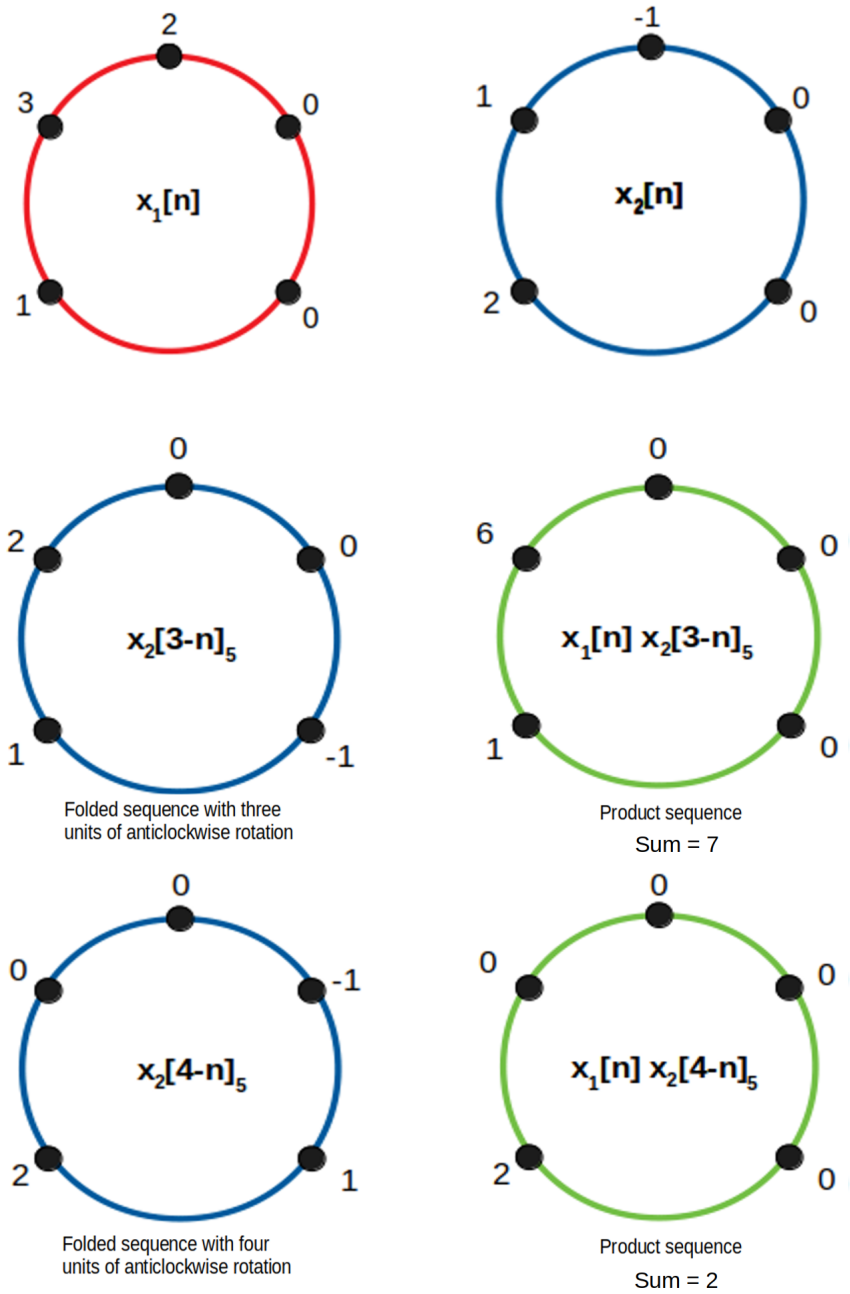
Therefore, linear convolution of $x[n]$ with $h[n]$ can be accomplished using circular convolution if the length of the sequences $x[n]$ and $h[n]$ are increased to N by zero padding. We illustrate this with the help of an example.

Example 1.1 *Consider the signals $x_1[n]$ and $x_2[n]$ given below. We want to find the linear convolution of the two signals using circular convolution.*

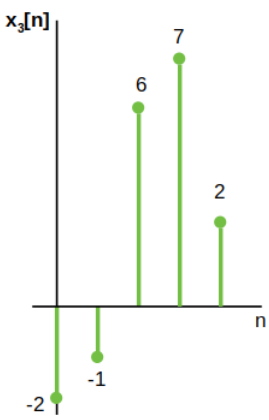


In order to compute the circular convolution, we look at each sequence as points on circles. Then using folding and rotation techniques, we compute the final result of the convolution. Refer to module-12b for a detailed explanation of how to find circular convolution.





Thus, $x_1[n] * x_2[n] =: x_3[n]$ *is given by*



The above example demonstrated the method of finding linear convolution using circular

convolution technique. The next example illustrates the methodology of using DFT and IDFT to find the response of an FIR filter.

Example 1.2 *Determine the response of the FIR filter with impulse response*[1] $h[n] = \{\underset{\uparrow}{1}, 2, 3\}$ *to the input sequence* $x[n] = \{\underset{\uparrow}{1}, 2, 2, 1\}$.

The input sequence has length $L = 4$ *and the impulse response has length* $M = 3$. *Linear convolution of these two sequences produces a sequence of length* $L + M - 1 = 6$. *Hence, the size of the DFTs must be at least* 6.

For simplicity we compute eight-point DFT ($N = 8$). *The reason for chosing an* 8*-point DFT will be clear in Module-12d. The eight point DFTs of* $x[n]$ *and* $h[n]$ *will be given by the following formulas:*

$$X[k] = \sum_{n=0}^{7} x[n] W_8^{kn} \qquad \textit{and} \qquad H[k] = \sum_{n=0}^{7} h[n] W_8^{kn}$$

On computation we get the following: The eight-pont IDFT is

k	$X[k]$	$H[k]$	$Y[k] = X[k]H[k]$
0	6	6	36
1	$\frac{2+\sqrt{2}}{2} - j\frac{4+3\sqrt{2}}{2}$	$1 + \sqrt{2} - j(3 + \sqrt{2})$	$-14.07 - j17.48$
2	$-1 - j$	$-2 - j2$	$j4$
3	$\frac{2-\sqrt{2}}{2} + j\frac{4-3\sqrt{2}}{2}$	$1 - \sqrt{2} + j(3 - \sqrt{2})$	$0.07 + j0.515$
4	0	2	0
5	$\frac{2-\sqrt{2}}{2} - j\frac{4-3\sqrt{2}}{2}$	$1 - \sqrt{2} - j(3 - \sqrt{2})$	$0.07 - j0.515$
6	$-1 + j$	$-2 + j2$	$-j4$
7	$\frac{2+\sqrt{2}}{2} + j\frac{4+3\sqrt{2}}{2}$	$1 + \sqrt{2} + j(3 + \sqrt{2})$	$-14.07 + j17.48$

$$y[n] = \frac{1}{8} \sum_{k=0}^{7} Y[k] W_8^{-kn}, \quad n = 0, 1, \cdots, 7 = \{\underset{\uparrow}{1}, 4, 9, 11, 8, 3, 0, 0\}.$$

Note that the first siz values of $y[n]$ *consitutes the desired output values. The last two values are zero because we used an eight-point DFT and IDFT, when, in fact, the minimum number of points required is six.*

What would have happened if instead of using eight-point DFT and IDFT we would have used a four-point DFT and IDFT. Check for yourself what the output would have been then.

[1] The symbol $\uparrow$ is used to denote the zero index of the sequence.